

Project Memory & Handoff Dossier: Artisan Furniture Studio

1. Project Overview & Current State

- **Project:** Production-ready e-commerce showroom & workshop management system for high-ticket solid wood furniture.
 - **Client / Brand:** `[CLIENT_NAME]` *Atelier* (Bespoke Solid Teak & Sheesham Wood).
 - **GitHub Repository:** `ShadowGeek2006/Furniture-Web-Service` (Branch: `main`).
 - **Deployment:** Vercel (`furniture-web-service.vercel.app`).
 - **Stack:** Next.js 14 (App Router, TypeScript, Tailwind CSS), Serverless PostgreSQL (Neon / Supabase), Meta WhatsApp Cloud API, Cloudinary (Image CDN).
-

2. Key Architecture & Operational Decisions

1. Zero-Friction Customer Experience:

- No customer login/passwords required. Guest cart stored in `localStorage`.
- Catalog browsing by Room (*Living, Dining, Bedroom, Study*) and Wood Variety (*Solid Teak, Sheesham, Oak*).
- Product detail pages with multi-angle photography, architectural dimensions blueprints (inches & cm), and timber seasoning specs.
- Guest checkout captures Name, WhatsApp Mobile, Delivery City, and Address.
- On checkout:
 - Server creates order with status `ENQUIRY_RECEIVED` and unique reference `ORD-YYYY-XXXX`.
 - Automated WhatsApp dispatch alerts the Shop Owner with customer details and itemized list.
 - Automated WhatsApp confirmation sent to customer acknowledging inquiry receipt.
 - On-screen fallback button opens pre-filled WhatsApp chat immediately.

2. Admin Operations & 1-Click GST Invoicing:

- Staff portal at `/admin/orders` displaying incoming customer inquiries, KPI metrics, and real-time search.
- Order adjustment workspace at `/admin/orders/[id]`:
 - Increment/decrement quantities with dynamic subtotal recalculation.
 - Remove items or add custom line items (carpentry surcharges, fabric upgrades, extra benches).

- Set negotiated trade discounts (₹) and white-glove delivery fees (₹).
 - Select GST calculation mode: *Inclusive*, *Exclusive (+18%)*, or *Exempt / Composition*.
 - Automatic Intra-State (CGST 9% + SGST 9%) vs Inter-State (IGST 18%) calculation based on customer state.
 - Record payment status (**PAID**, **PARTIAL**, **PENDING**) and method (**UPI**, **Bank Transfer**, **Cash**, **Cheque**).
 - **1-Click "Generate Official GST Bill"**: Locks order into an immutable, permanent invoice record (**INV-YYYY-XXXX**).
 - **Print-Ready A4 Tax Invoice (/admin/invoices/[id])**: Complete legal letterhead with HSN codes (**9403**), GST breakdown, 10-year warranty clause, print-optimized stylesheet for paper printing or instant "Save as PDF", and direct WhatsApp share button.
-

3. Codebase File Structure (46 Files)

- **Configuration:**
 - **package.json** (Next 14, React 18, Tailwind, TypeScript)
 - **tsconfig.json** (Strict paths @/*)
 - **tailwind.config.js** (Custom palette: **espresso**, **linen**, **brass**, **sand**)
 - **postcss.config.js** (Tailwind & Autoprefixer plugins)
 - **README.md** (Local setup, environment variables, Vercel deployment guide)
- **Design System & Atomic UI (src/components/ui/):**
 - **Button.tsx**, **Badge.tsx**, **Input.tsx**, **Textarea.tsx**, **Select.tsx**, **Card.tsx**, **Drawer.tsx**, **Modal.tsx**, **index.ts**
- **Global Layout (src/components/layout/):**
 - **Navbar.tsx** (Logo, category links, cart counter, WhatsApp connect)
 - **MobileNav.tsx** (Slide-out drawer with quick chat action)
 - **Footer.tsx** (Workshop story, showroom address, craft warranty)
- **Showroom & Homepage (src/components/home/):**
 - **HeroSection.tsx**, **RoomCategories.tsx**, **FeaturedCollection.tsx**, **CraftsmanshipStory.tsx**, **CustomFurnitureCTA.tsx**, **TrustFeatures.tsx**
- **Cart & Checkout (src/components/):**
 - **cart/CartDrawer.tsx** (Slide-out cart drawer)
 - **checkout/CheckoutModal.tsx** (Guest checkout form with validation and automated WhatsApp dispatch)
 - **product/ProductCard.tsx** (Responsive card with dimensions and quick add)
- **App Router Pages (src/app/):**
 - **globals.css** (Tailwind directives, fonts, print styles)

- `layout.tsx` (Root layout wrapping `CartProvider`, `Navbar`, `Footer`, `CartDrawer`)
 - `page.tsx` (Homepage assembling all editorial sections)
 - `shop/page.tsx` (Full catalog with dynamic search, filters, and sorting)
 - `shop/[slug]/page.tsx` (Product detail page with blueprint dimensions and finish selector)
 - `admin/layout.tsx` (Admin navigation bar)
 - `admin/orders/page.tsx` (Incoming orders workspace)
 - `admin/orders/[id]/page.tsx` (Order adjustment and 1-click GST invoice trigger)
 - `admin/invoices/page.tsx` (GST invoice ledger)
 - `admin/invoices/[id]/page.tsx` (Print-ready A4 tax invoice view)
 - **Backend API Routes (`src/app/api/`):**
 - `api/orders/route.ts` (Server-side price verification, order logging, automated WhatsApp dispatch)
 - `api/admin/invoices/route.ts` (Invoice retrieval and creation endpoints)
 - ****Data Contracts & Services (`src/types/`, `src/lib/`, `src/data/`):**
 - `types/product.ts`, `types/order.ts`, `types/invoice.ts`
 - `lib/utils.ts` (`formatINR()`, `formatDate()`, `cn()`)
 - `lib/CartContext.tsx` (Guest cart state in `localStorage`)
 - `lib/orderService.ts` (Order state management, seed data, GST math, invoice generation)
 - `lib/whatsappService.ts` (Meta WhatsApp Cloud API integration with fallback)
 - `data/mockProducts.ts` (Authentic solid wood furniture catalog data with specifications)
-

4. Issues Encountered & Resolved

1. Vercel Initial 404:

- *Cause:* Initial archive created via `shutil.make_archive` missed `src/app/page.tsx` and `src/app/layout.tsx` during directory traversal.
- *Resolution:* Rebuilt archive explicitly bundling all 45+ files; verified repository root structure.

2. TypeScript Compilation Error on `f.name`:

- *Error:* Type error: Argument of type `'string'` is not assignable to parameter of type `'SetStateAction<"Honey Teak" | ...>'`.
- *Cause:* In `src/app/shop/[slug]/page.tsx`, `selectedFinish` state was inferred strictly from `product.finish` union instead of `string`.
- *Resolution:* Explicitly typed state as `useState<string>(product.finish)` and `useState<string>(product.woodType)`.

3. Missing PostCSS Configuration:

- Added `postcss.config.js` with `tailwindcss` and `autoprefixer` to ensure smooth CSS compilation on Vercel.
-

5. Active Google Drive Artifacts

- **Project Documentation & Blueprint:** `Artisan Furniture Studio – Project Architecture & UI Design System` (Contains design tokens, data models, and component inventory)
 - **Current Patched Codebase Archive:** `artisan-furniture-studio-codebase.zip` (Contains all 46 verified production files ready for Vercel/GitHub)
-

6. Next Steps for Incoming Assistant / Next Chat

1. Verify latest commit deployment on Vercel at `furniture-web-service.vercel.app`.
2. Connect production Neon / Supabase PostgreSQL database using Prisma if transitioning from mock/localStorage to cloud persistence.
3. Configure live Meta WhatsApp Cloud API credentials in Vercel environment variables (`WHATSAPP_API_TOKEN`, `WHATSAPP_PHONE_NUMBER_ID`, `ADMIN_WHATSAPP_NUMBER`) when ready to send live WhatsApp messages to phone numbers.